

Asserten — Quickstart

asserten

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A Claude Code plugin for making LLM agents measurably more reliable.

Asserten takes any agent (its system prompt + tool definitions), audits it, generates a behavioral test suite, and shows you four versions side by side with the pass-rate on a fresh suite for each — then lets you gate deploys on the scenarios you care about.

version	what it is
v0	the prompt you brought, untouched
v1	v0 + the audit fixes you accepted
v2a	LIGHT optimization — pick-best across candidates (free, instant)
v2b	DEEP optimization — full cross-refine (~15 min)

Before you start: you need a backend

Asserten is a **thin client**. The audit, test generation, evaluation, and optimization all run in a separate **backend service** that holds the LLM API keys. The plugin just drives it over HTTP — so you never hand your Anthropic/OpenAI keys to the plugin.

You need a backend URL and (if it's enforced) an API key before anything will run. Ask whoever shared this with you for both.

1. Install (in Claude Code)

Marketplace (recommended):

```
/plugin marketplace add vinsing09/asserten  
/plugin install asserten@asserten
```

Or developer install:

```
git clone https://github.com/vinsing09/asserten ~/asserten
cd ~/asserten && pip install -e .
mkdir -p ~/.claude/plugins && ln -s "$PWD" ~/.claude/plugins/asserten
```

Restart Claude Code, then type `/asserten` — the commands should autocomplete.

2. Configure the backend

In the terminal where you launch Claude Code:

```
export ASSERTEN_BACKEND_URL=https://<your-backend-host> # default http://local-
host:8000
export ASSERTEN_API_KEY=<your-key> # if the backend en-
forces auth
```

If commands fail with a connection error, this is almost always the cause — the URL is unreachable or the key is missing.

3. Run the 10-minute flow

A sample agent (`examples/sample_agent.json`) ships with the plugin.

```
/asserten-ingest examples/sample_agent.json # hand over the agent → draft
/asserten-audit # see suggested prompt fixes
/asserten-select all # accept them → creates v0 + v1
/asserten-prepare-eval # generate the contract + test
cases
/asserten-eval v0 # eval the raw agent
/asserten-eval v1 # eval the audited agent
/asserten-compare # v0 vs v1 pass-rate table
```

That already answers “did the audit actually help?” To go further:

```
/asserten-optimize-deep <eval_run_id> # optimize v1 → v2b (~15 min)
/asserten-eval v2b
/asserten-scenarios v1 # ✓/x dashboard per scenario
/asserten-approve all-passing # lock in the passing scenarios
/asserten-deploy-gate {"candidate": "v2b"} # block a deploy that regresses
on them
```

Tip: the test-set size defaults to 40. Override it on `/asserten-prepare-eval {"count": 60}` — 15 is too small (it inflates the pass rate), and very large counts hit a redundancy ceiling.

Command cheat sheet

Setup — `/asserten (help)` · `/asserten-ingest <path>` · `/asserten-status` · `/asserten-reset`

Contract & tests — `/asserten-audit` · `/asserten-select all|1,3,5|none` · `/asserten-prepare-eval` · `/asserten-show-contract`

Your own test cases — `/asserten-byoe` (plain English) · `/asserten-add-tests <path>` · `/asserten-skip-tests <ids>` · `/asserten-unskip-tests <ids>`

Evaluate & optimize — `/asserten-eval <v0|v1|v2a|v2b>` · `/asserten-failures` · `/asserten-optimize-light` · `/asserten-optimize-deep <eval_run_id>` · `/asserten-compare`

Outcomes & deploy gate — `/asserten-scenarios <version>` · `/asserten-approve <ids|all-passing>` · `/asserten-unapprove <ids>` · `/asserten-deploy-gate {"candidate": "v2b"}`

One-shot — `/asserten-run <path>` (drives the whole flow, pausing at patch selection)

Bring your own agent

Instead of the sample, ingest your own. The shape is:

```
{
  "name": "My Agent",
  "raw_system_prompt": "You are ...",
  "tool_schemas": [{"name": "...", "description": "...", "parameters": { }}],
  "business_goal": "One line on what success means for this agent."
}
```

Save it as a `.json` file and `/asserten-ingest path/to/your_agent.json`.

Troubleshooting

symptom	fix
commands don't autocomplete	restart Claude Code; confirm the plugin installed
connection / timeout error	check <code>ASSERTEN_BACKEND_URL</code> is reachable
<code>401</code> / key error	set <code>ASSERTEN_API_KEY</code> to the key you were given
"no v1 yet"	run <code>/asserten-select</code> before <code>/asserten-prepare-eval</code>
eval says "generate contract first"	run <code>/asserten-prepare-eval</code> before <code>/asserten-eval</code>

Asserten is MIT-licensed and open: https://github.com/vinsing09/assertenSMARTPANTS_PRE-SERVED_215SMARTPANTS_PRESERVED_216